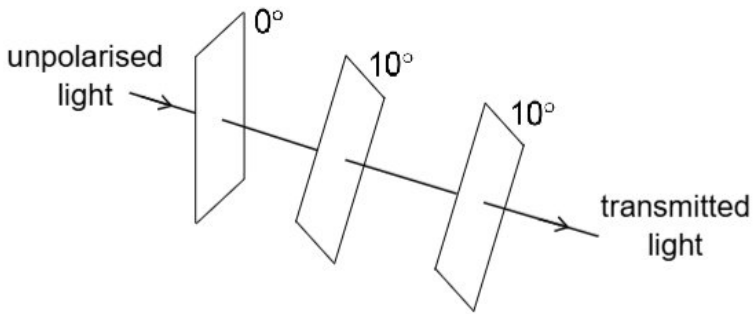
18 A narrow, parallel beam of unpolarised light is directed towards three ideal polarising filters. The beam meets the first filter with its axis of polarisation vertical. The axis of polarisation of the second filter is at an angle of 10° to the first filter. The third filter has its axis of polarisation parallel to the second filter as shown.



The third filter is now turned.
At what angle must the third filter be with respect to the second filter so that the intensity of the transmitted light is reduced to $\frac{1}{3}$ of the intensity of the unpolarised light?

A 34° **B** 44° **C** 47° **D** 54°